Re: [bascom] dmtf-chip mt8888 or mt8870 with AVR - timing problem

From "Bram Prosman" <bprosman@home.nl>
Date Sat, 6 Apr 2002 12:50:51 +0200

```
Hi Steve,

Made an application myself with the MT8880 and a 82S8252. Initially I had
the same problem, odd thing was that I had to remove the STK200 cable to get
interrupts working, attached a code snippit for the init routines which are
in production now and fully working. They are in Keil C though ( Sorry Mark
:-) ).

Regards, Bram


//**************************************************************************
***
//* Special register and control signal declaration
//**************************************************************************
***

sfr     DTMF =  0x80;           // Port P0 controlling MT8880
sbit    RS   =  P0^4;           // Register Select (11)
sbit    RW   =  P0^5;           // R/W (9)
sbit    CS   =  P0^6;           // Chip Select (10)
sbit    CLK  =  P0^7;           // Clk2 (12)


//**************************************************************************
***
//* Initialize MT8880
//**************************************************************************
***

void mt8880_init(void)

        {
        DTMF = 0xff;        // Force P0 to all outputs

        DTMF = 0x1c;        // Reg.A - B Reg. next,IRQ Ena.,CP Mode,Tone Off
        CLK = 1;            // Strobe into MT8880
        CLK = 0;

        DTMF = 0x11;        // Reg.B - 0, DTMF, NO Test, NO burst
        CLK = 1;            // Strobe into MT8880
        CLK = 0;
        }

//**************************************************************************
***
//* read_8880_status
//*
```

```
//* This routine reads the MT8880 status register and is used to clear
//* the interrupt flag.
//* We're not returning anything because we're not interested in the
contents
//* of the status register.
//*
***************************************************************************

void read_8880_status(void)

        {

        unsigned char dummy;

        RS = 1;                                 // Select status register
        RW = 1;                                 // Read mode
        CS = 0;                                 // Select MT8880
        CLK = 1;                                // Set clock high
        dummy = DTMF;                           // Read status register
        CLK = 0;                                // Set clock low
        CS = 1;                                 // Put MT8880 in tri-state
        RS = 0;                                 // Select data register

        }
```

----- Original Message -----
From: "Steve Childress" <stevech@san.rr.com>
To: <bascom@grote.net>
Sent: Saturday, April 06, 2002 05:56
Subject: RE: [bascom] dmtf-chip mt8888 or mt8870 with AVR - timing problem

> *I have the 8870 (DTMF decoder chip) working with an AVR with several*
> *compilers.*
> *For BASCOM, I had to resort to assembler in-line code to make it work. The*
> *BASCOM's built-ins wouldn't work.*
> *Here's a code segment...*
> *PORTD bit 6 is the 8870's OE signal*
> *PORTB's high 4 bits are the data lines from the 8870:*
>
```
>      sbi portd,6                                        ' dtmf chip
> output enable
>       Call Sleep(0)  ' probably not needed - a nop or two would do
>       in r24,22
>       cbi portd,6
>       lsr r24
>       lsr r24
>       lsr r24
>       lsr r24
>       sts {lastdtmf},r24
>       Lastdtmf = Lookup(lastdtmf , Dtmfnames)             ' make ASCII
```
>
> *steve*
>
>
> *-----Original Message-----*
> *From: owner-bascom@grote.net [mailto:owner-bascom@grote.net]On Behalf Of*
> *Michael*
> *Sent: Friday, April 05, 2002 5:18 AM*

> *To: bascom@grote.net*
> *Subject: [bascom] dmtf-chip mt8888 or mt8870 with AVR - timing problem*
>
>
>
>
> *Hello folks,*
>
> *I have some troubvle with the timing of MT8888 DTMF-Transceiver and MT8870*
> *Receiver and AVR. Has anyone written a bascom-avr-code to handle the*
chips?
> *It would help me, because I get no Interrupts from them and I think, the*
> *reason can be my initialisation.*
>
> *best regards*
> *Michael*
>
> *mwoeste@web.de*
>
>
>
>
>